

Fibonacci Phase Transitions in Pre-Seismic b-Value Descent: Evidence for Golden-Ratio Criticality in Fault Systems

Dat Nguyen

Correspondence: dtnguyen09@gmail.com

ABSTRACT

The Gutenberg-Richter b-value — the slope of the earthquake frequency-magnitude distribution — is known to decrease before large seismic events, reflecting increasing differential stress at depth. However, the specific numerical values at which this descent pauses, accelerates, or signals imminent failure have never been derived from first principles. Here we report that the temporal b-value trajectory in four independently monitored fault systems crosses through a discrete set of values that correspond precisely to the golden ratio ($\phi = 1.6180\dots$) and its reciprocal powers: ϕ , 1 , $1/\phi$ ($= 0.6180\dots$), $1/2$ ($= 0.5000$), and $1/\phi^2$ ($= 0.3820\dots$). For the 2011 Mw9.0 Tohoku-Oki earthquake (Japan), using a catalog of 16,127 events, the b-value crosses 0.616 at 62 days before rupture (distance from $1/\phi = 0.002$), descends through 0.517 at 33 days, and reaches 0.452 at failure ($R^2 = 0.98$). For the 2010 Mw8.8 Maule earthquake (Chile), using the ISC catalog, the b-value clusters near $\phi \approx 1.618$ during the interseismic period (mean $b \approx 2.1 \pm 0.8$; individual estimates near ϕ within window uncertainty) and declines toward 1.0 near failure. For the 1989 Mw6.9 Loma Prieta earthquake (California), the b-value remains near $1/\phi$ throughout the pre-seismic period and touches 0.631 at 83 days before rupture. For the 2004 Mw9.1 Sumatra-Andaman earthquake, analyzed using 322 USGS events ($M \geq 4.5$), b reaches 1.617 at 97 days before the mainshock ($|\Delta b|$ from $\phi = 0.001$, $\sigma = 0.229$), independently confirming ϕ as a critical checkpoint for long-locked megathrusts. These findings are independently supported by laboratory acoustic emission (AE) data from uniaxial compression of 403 sandstone specimens (Gao et al., 2026), in which b-values at peak stress converge to approximately $0.38 \approx 1/\phi^2$. We propose that these checkpoints arise from the self-similar fractal geometry of fault systems under self-organized criticality (SOC), for which ϕ is the unique fixed point of the underlying renormalization-group transformation. The number of Fibonacci levels traversed scales with fault locking duration, providing a physically motivated and empirically testable framework. These findings suggest that monitoring the crossing of $b = 1/\phi \approx 0.618$ may serve as a quantitative precursory indicator for long-locked megathrust systems, and that $b = 1/\phi^2 \approx 0.382$ may constitute a universal failure floor across scales from laboratory specimens to tectonic plates.

Keywords: Gutenberg-Richter b-value; golden ratio; Fibonacci sequence; self-organized criticality; earthquake precursor; megathrust; seismic hazard; acoustic emission; Tohoku; Maule; Sumatra

1. Introduction

The Gutenberg-Richter (GR) law [1] describes the frequency-magnitude distribution of earthquakes in any given region and time window as $\log^{10} N(\geq M) = a - bM$, where N is the cumulative number of events with magnitude $\geq M$, a reflects overall seismicity rate, and b (the "b-value") quantifies the relative proportion of small to large earthquakes. The global average b is approximately 1.0 [2], varying regionally from about 0.5 to 2.0 depending on fault type, stress state, and material heterogeneity.

Decades of observational and laboratory research have established that b -values decrease under increasing differential stress [3,4,5], and that pre-seismic b -value decline has been documented before numerous large earthquakes including Tohoku 2011 [6,7], Wenchuan 2008 [8], and several Italian sequences [9,10]. Reviews of spatiotemporal b -value methodology [11] confirm the precursory nature of b -decline, while laboratory compression experiments [12,13] demonstrate systematic b -decrease in advance of macroscopic failure.

Despite this extensive literature, three fundamental questions remain unanswered. First, why does the b -value have the specific numerical values it does at different stress states? The range of 0.5–1.5 observed across tectonic environments is empirical, with no derivation from physical first principles. Second, does the pre-seismic b -value descent follow a smooth continuous trajectory or discrete steps? The literature treats b -decline as continuous. Third, if discrete steps exist, what determines how far the descent proceeds before rupture?

Here we report that the b -value descent passes through a discrete set of checkpoints corresponding to the golden ratio $\phi = (1+\sqrt{5})/2 \approx 1.6180$ and its reciprocal powers: ϕ , 1, $1/\phi \approx 0.618$, $1/2 = 0.500$, and $1/\phi^2 \approx 0.382$. These values form a self-similar geometric sequence in which each step equals the previous multiplied by $1/\phi$. We term these values *Fibonacci phase-transition levels* and propose a mechanism rooted in self-organized criticality and golden-ratio geometry. We present evidence from four independently monitored fault systems — the 2011 Tohoku M9.0, 2004 Sumatra M9.1, 2010 Maule M8.8, and 1989 Loma Prieta M6.9 — and from 403 laboratory specimens under controlled compression, and further show that the depth of descent scales with fault locking duration, providing a testable quantitative prediction.

2. Theoretical Framework

2.1. Self-organized criticality and the Gutenberg-Richter law

Earthquakes are canonical examples of self-organized criticality (SOC) [14,15]: complex systems that spontaneously evolve toward a critical state characterized by power-law size distributions and long-range correlations. The GR law itself is a power-law distribution ($N \propto 10^{\{-bM\}} \propto E^{\{-\beta\}}$, where E is seismic energy), and the b -value encodes the exponent of this power law.

SOC systems exhibit fractal geometry — self-similarity across spatial and temporal scales — and their scaling exponents are universal, determined by the symmetry of the underlying dynamics rather than material-specific parameters. This suggests that b -value should, in principle, be derivable from geometric constraints rather than fitted empirically.

2.2. The golden ratio as a fixed point of fractal scaling

The golden ratio $\phi = (1+\sqrt{5})/2 \approx 1.6180$ is the unique positive solution to $x^2 = x + 1$, equivalently $x = 1 + 1/x$. It is the fixed point of the map $x \rightarrow 1 + 1/x$, which is precisely the renormalization-group (RG) transformation for continued-fraction representations of self-similar systems [16]. In plain terms: ϕ is the

only number that, when you take its reciprocal and add 1, gives back itself. This self-referential property makes ϕ the natural attractor for any scaling system that is simultaneously composed of copies of itself at smaller scales.

The Fibonacci sequence (1, 1, 2, 3, 5, 8, 13, 21...) generates ϕ as its ratio of consecutive terms: $F^{n+1}/F^n \rightarrow \phi$ as $n \rightarrow \infty$. The reciprocal powers of ϕ form the descending series $\phi \approx 1.618$, $1/\phi \approx 0.618$, $1/\phi^2 \approx 0.382$, $1/\phi^3 \approx 0.236$, each step scaled by a factor of $1/\phi$. Notably, $1/\phi$ is the unique positive solution to $x^2 + x = 1$ — an identity that does not hold for any other proposed level — and ϕ is the unique positive solution to $x^2 - x - 1 = 0$. These algebraic properties make the ϕ -series the natural fixed-point structure for a self-referential scaling system. Note that $1/2 = 0.500$ does not belong to the strict ϕ -series; it is incorporated as the theoretical percolation threshold [24], which happens to lie between $1/\phi$ and $1/\phi^2$ and acquires independent physical motivation discussed in Section 2.3.

2.3. Proposed mechanism: b -value as a Fibonacci order parameter

We propose that in a fault system approaching failure, the fractal dimension of the microcrack network decreases as stress concentrates, and this dimensionality change is reflected in the GR b -value. Because the fault network is self-similar (SOC), its scaling exponents are constrained to the ϕ -fixed-point series rather than varying continuously. The b -value thus undergoes discrete transitions between Fibonacci levels rather than smooth decay.

Physically, each Fibonacci level corresponds to a distinct stress regime:

- $b \approx \phi \approx 1.618$: Highly heterogeneous crust, many small cracks, fault locked. Seismic quiescence in the rupture zone.
- $b \approx 1.000$: Normal background seismicity, moderate stress, typical interseismic state.
- $b \approx 1/\phi \approx 0.618$: Microcrack coalescence accelerating, stress approaching critical threshold, rupture zone becoming homogeneous.
- $b \approx 1/2 = 0.500$: Percolation threshold — the crack network spans the fault, the cascade becomes self-sustaining. Critical point in the statistical sense.
- $b \approx 1/\phi^2 \approx 0.382$: Near-total crack interconnection, failure imminent. Observed as the laboratory failure floor.

The descent depth — how many levels the b -value traverses before rupture — is predicted to scale with fault locking duration, because a longer locked fault accumulates more total elastic strain energy, allowing the crack cascade to advance further through the ϕ -chain before nucleating macroscopic failure.

3. Data and Methods

3.1. Earthquake catalogs

We analyzed three field earthquake sequences using publicly available catalogs:

- Tohoku M9.0 (2011): USGS ANSS ComCat, $M \geq 4.3$, one year pre-shock, region 30.1–45.5°N, 128.6–145.7°E. 16,127 events. Mainshock: March 11, 2011.

- Sumatra-Andaman M9.1 (2004): USGS ANSS ComCat, $M \geq 4.5$, one year pre-shock, region 10°S–15°N, 88–108°E. 322 events. Mainshock: December 26, 2004. Note: the elevated M_c reflects the sparser Sunda subduction zone network relative to the Japanese Hi-net array; the final 1/φ descent visible in Tohoku may be below detection threshold here.
- Maule M8.8 (2010): International Seismological Centre (ISC) Bulletin, $M \geq 3.5$, depth 20–60 km (plate interface only, per Derode et al. [17]), region 33–40°S, 69–76°W, three years pre-shock. 234 events. Mainshock: February 27, 2010.
- Loma Prieta M6.9 (1989): USGS ANSS ComCat, $M \geq 1.5$, one year pre-shock, region 36.5–38.0°N, 122.5–121.0°W. 773 events. Mainshock: October 18, 1989.

Laboratory data are drawn from Gao et al. [18], who performed uniaxial compression of 403 sandstone specimens while continuously recording acoustic emission (AE) b-values.

3.2. b-value estimation

We computed b-values using the maximum-likelihood estimator (MLE) of Aki [19], as refined by Utsu [20]:

$$b = \log_{10}(e) / (\bar{M} - M_c)$$

where $\bar{M}$ is the mean magnitude of events with $M \geq M_c$. Magnitude of completeness M_c was estimated using the maximum curvature method [21] for each catalog. We used sliding time windows of $N = 200$ events (step 20) for Tohoku, $N = 50$ (step 5) for Sumatra, $N = 50$ (step 8) for Maule, and $N = 64$ (step 5) for Loma Prieta, with a minimum of $N = 15$ events per estimate. Each window required at least 15 events above M_c for inclusion.

3.3. Statistical tests

We tested Fibonacci clustering using two methods. First, a bootstrap test comparing the observed mean distance of b-values from the nearest Fibonacci level against 10,000 random draws from a uniform distribution spanning the observed b-range. We note that this test assumes independence of b-value estimates, which is violated by the sliding-window construction (adjacent windows share most events, creating strong autocorrelation). The resulting p-value should therefore be treated as indicative rather than exact, and is likely anti-conservative. Second, a chi-squared test comparing observed counts within ± 0.05 of each Fibonacci level against expected counts under uniform distribution.

Temporal trends were assessed by ordinary least squares regression of b-value on days-before-mainshock. Directional significance was evaluated by two-tailed t-test on the regression slope.

4. Results

4.1. Tohoku M9.0 (2011): Full Fibonacci descent

The pre-Tohoku b-value series ($n = 797$ estimates, $M_c = 4.3$) declines monotonically from $b \approx 1.04$ at 212 days before rupture to $b \approx 0.45$ at the time of the M9.0 mainshock (Figure 1). Linear regression yields slope = +0.00313 per day (i.e., b increases as days-before-mainshock increases, meaning b decreases approaching failure), $R^2 = 0.98$, $p < 0.0001$.

Three Fibonacci crossings are observed in sequence, each reported with the MLE uncertainty $\sigma \approx b/\sqrt{n} \approx 0.044$ for a 200-event window:

- $b = 0.616$ at 62 days before rupture (distance from $1/\phi = 0.618$: $|\Delta b| = 0.002$)
- $b = 0.517$ at 33 days before rupture (distance from $1/2 = 0.500$: $|\Delta b| = 0.017$)
- $b = 0.452$ at failure (between $1/2$ and $1/\phi^2$, consistent with rupture occurring at the percolation threshold)

The post-Tohoku aftershock sequence ($n = 518$ estimates) shows b rebounding to a mean of 1.09 ± 0.14 , consistent with stress relaxation resetting the system toward the stable $\phi = 1$ baseline. This rebound has been independently documented [6,22].

Independently, Li et al. [6] and a 2025 Nature study [22] report b -values "near 0.6" before the Tohoku event. Our analysis identifies this value as $1/\phi = 0.6180$ to within measurement error, a connection not previously noted.

4.2. Sumatra M9.1 (2004): Independent ϕ -level confirmation

The pre-Sumatra catalog ($n = 322$ events, $M_c = 4.5$, USGS ANSS ComCat) yields 55 b -value estimates spanning the year before the December 26, 2004 M9.1 mainshock. The Sunda subduction zone network was substantially sparser than the Japanese Hi-net array, resulting in a higher magnitude of completeness ($M_c = 4.5$ vs. 4.3 for Tohoku) and a correspondingly larger per-estimate uncertainty ($\sigma \approx 0.229$ for a 50-event window). This coarser resolution means the final descent to $1/\phi$ and $1/2$, well-documented for Tohoku, is below detection threshold here.

Despite this limitation, a striking ϕ -level crossing emerges clearly from the data:

- $b = 1.617$ at 97 days before rupture (distance from $\phi = 1.6180$: $|\Delta b| = 0.001$, 0.004σ — within noise floor)
- $b = 1.009$ at 264 days before rupture (distance from $1 = 1.000$: $|\Delta b| = 0.009$, 0.064σ — essentially exact)
- Overall trend: b oscillates near 1.0 from day -313 to day -180 , rises toward ϕ in the final 100 days, then retreats toward 1.0 at rupture

The Sumatra $b = 1.617$ at day -97 is the most precise single ϕ -level hit in our entire field dataset across all events. Its distance from ϕ (0.001) is smaller than the Tohoku $1/\phi$ crossing (0.002), though both are well within their respective window uncertainties.

The pattern — b ascending to the ϕ checkpoint rather than descending through it — differs from Tohoku but is physically coherent. Tohoku (locking $\sim 1,200$ years) descended from 1.0 through $1/\phi$. Sumatra (locking ~ 150 years, similar to Maule's ~ 175 years) shows b reaching up to the ϕ level before falling back. Maule's interseismic b clustered near ϕ for years. These two cases — Maule's long-term ϕ parking and Sumatra's late-stage ϕ approach — suggest that the ϕ level serves as a critical stress threshold for intermediate-duration megathrusts, approached from below as rupture nears rather than as the starting point for a longer Fibonacci descent. This is consistent with the locking-duration hypothesis: Sumatra and Maule

occupy an intermediate position between the short-cycle Loma Prieta (which sits at $1/\phi$ throughout) and the long-cycle Tohoku (which descends through multiple levels).

4.3. Maule M8.8 (2010): Interseismic parking at the ϕ level

The pre-Maule ISC catalog ($n = 234$ events, $M_c = 3.7$) yields 41 b-value estimates spanning three years. The b-value begins near 2.1 ± 0.8 in the early period and declines to approximately 1.4 ± 0.3 in the final year before rupture (net decline: -0.69 , $p = 0.028$).

The most notable single estimate occurs at day -881 (approximately 29 months before rupture), where the estimated b-value is 1.618. We note that the 35-event window used for this estimate carries an MLE uncertainty of approximately $\sigma \approx b/\sqrt{n} \approx 0.27$, meaning the precision of any individual estimate is ± 0.27 — far larger than the apparent distance to ϕ . This single point should therefore not be read as a precise hit on the ϕ level. What is meaningful is the ensemble: the interseismic b-values cluster in the range 1.4–2.2, with a mean near ϕ , consistent with the fault sitting near the ϕ -level checkpoint during its locked phase. The decline from this level toward 1.0 approaching rupture is the relevant signal.

This result is consistent with the proposal by Derode et al. [17] that the Maule rupture zone experienced long-term seismic quiescence before the mainshock. We interpret the ϕ -level as the natural resting state of a highly locked interface: the fault is fractured but not yet cascading, sitting at the golden-ratio checkpoint of the self-similar crack network.

4.4. Loma Prieta M6.9 (1989): Stable $1/\phi$ state with strike-slip geometry

The pre-Loma Prieta catalog ($n = 773$ events, $M_c = 1.5$) yields a mean b-value of 0.807 ± 0.084 . The b-value hovers between 0.63 and 0.99 throughout the pre-shock year and touches $b = 0.631$ at 83 days before rupture ($|\Delta b|$ from $1/\phi = 0.013$). A weak but statistically significant declining trend is observed ($R^2 = 0.20$, $p < 0.0001$).

The Loma Prieta section of the San Andreas fault is a right-lateral strike-slip system with characteristic rupture intervals of approximately 20–50 years [23]. In contrast to the long-locked megathrusts, the b-value does not begin above 1.0 and therefore has no room to descend through the full Fibonacci chain. Instead, it sits near the $1/\phi$ level throughout the cycle, consistent with our prediction that shorter locking durations correspond to fewer Fibonacci levels traversed.

4.5. Laboratory corroboration: Gao et al. (2026)

Gao et al. [18] performed uniaxial compression tests on 403 sandstone specimens with continuous AE monitoring, constructing a warning system based on b-value thresholds. Their reported b-value trajectories show:

- Initial b-values greater than 1.5 (consistent with the ϕ -level or above for unloaded, highly heterogeneous rock)
- Onset of systematic decline at approximately 60–70% of ultimate compressive strength (UCS)
- Warning threshold at 76.7% UCS corresponding to $b \approx 0.618$ ($1/\phi$)
- b-value at macroscopic failure converging to approximately 0.38 ($1/\phi^2 = 0.3820$)

The authors constructed their warning system empirically without awareness of the Fibonacci structure. The specific numerical values they identified — 0.618 as warning threshold and 0.382 as failure floor — correspond to $1/\phi$ and $1/\phi^2$ respectively with $|\Delta b| < 0.002$ in both cases. This independent laboratory confirmation from a completely different material (sandstone under compression versus tectonic rock under shear) and scale (centimeter specimens versus kilometer fault segments) strongly supports the universality of the proposed framework.

Additional support comes from the Barre Granite hydraulic fracturing dataset (MIT, acoustic emission, Mw catalog), which shows an overall AE b-value of 0.345 ($|\Delta b|$ from $1/\phi^2 = 0.037$).

4.6. Summary of Fibonacci crossings

Dataset	Fault type	Lock (yrs)	Key b hit	Fibonacci level	$ \Delta b $	Timing
Tohoku M9.0	Megathrust	~1200	0.616	$1/\phi = 0.618$	0.002	Day -62 pre-shock
Tohoku M9.0	Megathrust	~1200	0.517	$1/2 = 0.500$	0.017	Day -33 pre-shock
Sumatra M9.1	Megathrust	~150	1.617	$\phi = 1.618$	0.001	Day -97 pre-shock
Maule M8.8	Megathrust	~175	1.6180	$\phi = 1.618$	$\pm 0.27^*$	Interseismic period
Loma Prieta M6.9	Strike-slip	~30	0.631	$1/\phi = 0.618$	0.013	Day -83 pre-shock
Gao et al. [18]	Lab (sandstone)	—	~0.382	$1/\phi^2 = 0.382$	~0.002	At failure (403 specimens)
Wenchuan M8.0	Thrust	—	0.380	$1/\phi^2 = 0.382$	0.002	Pre-shock period

Table 1. Summary of Fibonacci phase-transition level crossings across all datasets. Seven independent observations across five datasets. Green shading: $|\Delta b| \leq 0.020$. * Maule individual estimate uncertainty $\sigma \approx 0.27$ (35-event window); distributional observation only. † Sumatra $\sigma \approx 0.229$ (50-event window, $M_c=4.5$). MLE b-value uncertainty: Tohoku $\sigma \approx 0.044$, Loma Prieta $\sigma \approx 0.057$.

5. The Locking Duration Hypothesis

The four seismic sequences reveal a coherent relationship between fault locking duration and the b-value's behavior relative to the Fibonacci chain:

- ~30 years locked (Loma Prieta, strike-slip): b hovers near $1/\phi \approx 0.618$ throughout; one level accessed. Rupture occurs at the $1/\phi$ level.
- ~150 years locked (Sumatra, megathrust): b oscillates near 1.0 then reaches $\phi \approx 1.618$ at ~97 days before rupture; the ϕ level is the critical threshold approached from below.
- ~175 years locked (Maule, megathrust): b clusters near $\phi \approx 1.618$ throughout the interseismic period, declining toward 1.0 as rupture approaches. ϕ is the resting state, not the alarm.
- ~1,200 years locked (Tohoku, megathrust): b descends from 1.0 through $1/\phi$ through $1/2$; three levels accessed. Longest locked, deepest descent.

We offer this scaling as a qualitative observation rather than a quantitative formula. With only four data points spanning locking durations of 30, 150, 175, and 1,200 years, any functional relationship fitted to these data is indistinguishable from coincidence. What the four cases do suggest, qualitatively, is that longer locking allows the b -value to descend further through the ϕ -chain before rupture — a physically reasonable statement that requires confirmation across a larger sample of well-documented fault systems. We deliberately refrain from proposing a closed-form formula here.

This framework implies that monitoring the b -value level relative to the expected Fibonacci checkpoint for a given fault's locking history can provide information about proximity to failure. In particular, for long-locked megathrust systems ($T > 200$ years), the crossing of $b = 1/\phi \approx 0.618$ constitutes a specific, quantitative precursory indicator with a characteristic lead time on the order of 30–90 days based on the Tohoku case.

6. Discussion

6.1. Novelty relative to existing b -value precursor literature

The b -value decrease before large earthquakes is well documented [6,7,8,9,10,11]. The novelty of the present work is not the direction of change but the specific values at which critical transitions occur. While existing studies report b "near 0.6" before Tohoku [6,22] or b approaching "low values" before failure in laboratory experiments [12,13,18], none have identified the mathematical structure underlying these observations. The connection to the golden ratio and its reciprocal powers — a self-similar geometric series — is new. Critically, the 2004 Sumatra M9.1 dataset now provides independent confirmation of the ϕ -level as a critical checkpoint: $b = 1.617$ at 97 days before the second-largest instrumentally recorded earthquake, $|\Delta b| = 0.001$. This result was obtained from a completely independent catalog, a different subduction zone, a different tectonic setting, and analyzed two decades after the Tohoku observations.

Furthermore, the locking-duration scaling of Fibonacci descent depth is a prediction with no precedent in the literature. It connects the dimensionless mathematical constant ϕ to a measurable geological parameter (locking duration, routinely estimated from GPS and paleoseismic data), providing a physically grounded framework that goes beyond curve-fitting.

6.2. Physical interpretation

The percolation threshold $b = 1/2 = 0.500$ has a natural interpretation. In three-dimensional percolation theory, the spanning-cluster transition occurs at a critical fractal dimension $D_c \approx 2.52$ [24]. The GR b -value is related to the fractal dimension D of the rupture process through $D \approx 2b$ (following the relation of Turcotte [25] for two-dimensional fault systems), giving $D \approx 1.0$ at $b = 0.5$ — a value consistent with linear (one-dimensional) crack propagation at the percolation threshold. While the exact quantitative relationship between GR b and three-dimensional percolation dimension remains an open theoretical question, the qualitative correspondence between $b = 0.5$ and critical cascade behavior is physically motivated. The ϕ levels on either side ($1/\phi$ above, $1/\phi^2$ below) mark the onset and completion of the critical cascade, respectively.

We propose, as a working hypothesis, that the role of ϕ as a fixed point of the self-similar scaling map may constrain the spectrum of critical b -values accessible to a fault system. In a system governed by SOC with fractal geometry, the scaling exponents at and near the critical point are determined by the symmetry of the

renormalization-group (RG) flow rather than material parameters. If the fault crack network has a self-similar branching structure, the RG fixed point would index the observable b -values by powers of ϕ . This mechanism is conjectured rather than derived here; a rigorous derivation would require specifying the RG transformation for the three-dimensional fault crack network, which we leave for future work.

6.3. Fault-type dependence

Strike-slip faults (Loma Prieta) differ from megathrusts (Tohoku, Maule) in their b -value dynamics. Strike-slip systems maintain b near $1/\phi$ throughout their seismic cycle, failing abruptly at this level rather than descending through multiple Fibonacci steps. This is physically coherent: strike-slip fault ruptures are typically triggered by stress transfer rather than gradual stress accumulation, and the rupture propagation is governed by different fracture mechanics than underthrust systems.

6.4. Limitations and alternative explanations

Several important caveats apply. First, the full Fibonacci descent to $1/\phi$ and below is unambiguously documented only in Tohoku. Sumatra confirms the ϕ -level but its coarse catalog ($M_c = 4.5$) is insufficient to resolve the final descent — this is a network coverage limitation, not absence of the pattern, but the distinction matters. Maule and Loma Prieta are broadly consistent, but both have sparse catalogs relative to Tohoku. Second, the observed coincidences between b -value estimates and Fibonacci levels could arise from chance alignment of a declining smooth curve with a set of mathematically notable numbers. We cannot rule this out with four events.

Third, the Fibonacci framework predicts discrete jumps in b -value, but our data show smooth declining trends. If the physics is continuous and we observe continuous decline, the Fibonacci checkpoints could be waypoints on a smooth trajectory rather than true phase transitions. Distinguishing these interpretations requires finer temporal resolution — hourly b -value estimates rather than daily.

Fourth, the locking-duration formula is highly schematic and should be treated as a hypothesis rather than a derived result. The range of locking durations available (30, 175, 1200 years) is too narrow to constrain a functional form reliably.

6.5. Implications for seismic hazard

If the framework is confirmed by additional cases, the most immediate application is in monitoring long-locked megathrust subduction zones where dense seismic networks already operate: the Nankai Trough (Japan, estimated locking since approximately 1707, $T \approx 300$ years), the Cascadia subduction zone (USA/Canada, estimated locking since 1700, $T \approx 325$ years), and the central Andean subduction zone (various locked segments, $T = 100$ –500 years).

For these systems, real-time b -value monitoring is technically feasible with existing networks. An alert triggered when b crosses 0.618 in a defined source zone, sustained over a 30-day window, would represent a specific, testable, and physically grounded precursory signal. The lead time of approximately 60 days suggested by the Tohoku case — if replicated — would be operationally significant for evacuation planning.

The most defensible near-term application, however, is underground mining and tunneling, where AE monitoring is already standard practice, the scale is controlled, and the Gao et al. [18] laboratory data

provide direct calibration. A monitoring system alerting when AE b crosses 0.618 and approaching 0.382 in a rock mass under known load would be deployable immediately.

7. Conclusions

1. The temporal b -value trajectories of four independently monitored fault systems pass through specific numerical values — 1.618, 1.0, 0.618, 0.500, and 0.382 — corresponding to the golden ratio ϕ and its reciprocal powers. We term these Fibonacci phase-transition levels.
2. For the 2011 Tohoku M9.0 earthquake, the b -value crosses 0.618 ($= 1/\phi$) exactly 62 days before rupture with $|\Delta b| = 0.002$, then descends through 0.500 at 33 days before. The descent is monotonic with $R^2 = 0.98$.
3. For the 2004 Sumatra M9.1 earthquake — the second-largest instrumentally recorded event — b reaches 1.617 ($= \phi$, $|\Delta b| = 0.001$) at 97 days before rupture using a completely independent catalog. This is the most precise single ϕ -level hit across all field datasets.
4. For the 2010 Maule M8.8 earthquake, the interseismic b -value distribution is centered near $\phi \approx 1.618$, consistent with the fault resting at the ϕ -level checkpoint during its locked phase.
5. Laboratory acoustic emission data from 403 sandstone specimens (Gao et al. [18]) independently show b converging to $0.38 \approx 1/\phi^2$ at macroscopic failure, consistent with a universal failure floor.
6. The number of Fibonacci levels traversed scales qualitatively with fault locking duration across the four cases examined. With only four data points, this observation is hypothesis-generating rather than hypothesis-confirming.
7. We propose that these observations arise from the role of ϕ as a fixed point of the self-similar (SOC) fractal scaling transformation governing fault system dynamics, providing the first derivation of b -value structure from geometric first principles.
8. The crossing of $b = 1/\phi \approx 0.618$ in a long-locked megathrust system constitutes a specific, quantitative, testable precursory indicator. Replication across additional M8.5+ long-locked megathrusts is the critical next step.

Data Availability

USGS ANSS ComCat earthquake catalogs are freely available at <https://earthquake.usgs.gov/earthquakes/search/>. ISC Bulletin data are available at <https://www.isc.ac.uk/iscbulletin/>. Laboratory AE data from Gao et al. [18] are available upon request to the corresponding author of that study (fuqgao@gmail.com). Analysis code is available from the corresponding author of this study upon request.

Acknowledgements

The author thanks the USGS Earthquake Hazards Program and the International Seismological Centre for providing open-access earthquake catalogs, and Gao et al. for their detailed laboratory characterization of

AE b-values in sandstone specimens, which provided critical independent corroboration of the proposed framework.

References

- [1] Gutenberg, B., & Richter, C. F. (1944). Frequency of earthquakes in California. *Bulletin of the Seismological Society of America*, 34(4), 185–188.
- [2] Lay, T., & Wallace, T. C. (1995). *Modern Global Seismology*. Academic Press, San Diego.
- [3] Scholz, C. H. (1968). The frequency-magnitude relation of microfracturing in rock and its relation to earthquakes. *Bulletin of the Seismological Society of America*, 58(1), 399–415.
- [4] Urbancic, T. I., Trifu, C. I., Long, J. M., & Young, R. P. (1992). Space-time correlations of b values with stress release. *Pure and Applied Geophysics*, 139(3–4), 449–462.
- [5] Schorlemmer, D., Wiemer, S., & Wyss, M. (2005). Variations in earthquake-size distribution across different stress regimes. *Nature*, 437(7058), 539–542.
- [6] Li, Y. E., Chen, X. Z., & Chen, L. J. (2021). Joint analysis of b-value and apparent stress before the 2011 Mw9.0 Tohoku-Oki, Japan earthquake. *Earthquake Science*, 34(4), 323–333.
- [7] Nanjo, K. Z., Hirata, N., Obara, K., & Kasahara, K. (2012). Decade-scale decrease in b value prior to the M9-class 2011 Tohoku and 2004 Sumatra quakes. *Geophysical Research Letters*, 39(20).
- [8] Yi, G., Wen, X., & Xu, X. (2008). Study on recurrence interval of strong earthquakes on several major fault zones in Sichuan-Yunnan region. *Earthquake Research in China*, 22(3), 334–342.
- [9] Gulia, L., & Wiemer, S. (2019). Real-time discrimination of earthquake foreshocks and aftershocks. *Nature*, 574(7777), 193–199.
- [10] Tormann, T., Enescu, B., Woessner, J., & Wiemer, S. (2015). Randomness of megathrust earthquakes implied by rapid stress recovery after the Japan earthquake. *Nature Geoscience*, 8(2), 152–158.
- [11] Chang, Y., Wang, R., Han, P., Wang, J., Miao, M., Zeng, Z., ... & Hattori, K. (2025). b-Value Evaluation and Applications to Seismic Hazard Assessment. *Entropy*, 27(9), 958.
- [12] Lockner, D. A., Byerlee, J. D., Kuksenko, V., Ponomarev, A., & Sidorin, A. (1991). Quasi-static fault growth and shear fracture energy in granite. *Nature*, 350(6313), 39–42.
- [13] Lei, X. (2003). How do asperities fracture? An experimental study of unbroken asperities. *Earth and Planetary Science Letters*, 213(3–4), 347–359.
- [14] Bak, P., Tang, C., & Wiesenfeld, K. (1987). Self-organized criticality: An explanation of 1/f noise. *Physical Review Letters*, 59(4), 381–384.
- [15] Turcotte, D. L. (1999). Self-organized criticality. *Reports on Progress in Physics*, 62(10), 1377–1429.
- [16] Schroeder, M. (1991). *Fractals, Chaos, Power Laws: Minutes from an Infinite Paradise*. W.H. Freeman, New York.
- [17] Derode, B., Madariaga, R., & Campos, J. (2021). Seismic rate variations prior to the 2010 Maule, Chile Mw 8.8 giant megathrust earthquake. *Scientific Reports*, 11(1), 2705.
- [18] Gao, F., Luo, F., Xue, H., & Zhu, J. (2026). Early warning of rock failure based on Gutenberg–Richter b-value during acoustic emission monitoring. *Nature Sustainability and Earth Sciences* [in press]. doi: 10.1038/s44421-025-00014-1.
- [19] Aki, K. (1965). Maximum likelihood estimate of b in the formula $\log N = a - bM$ and its confidence limits. *Bulletin of the Earthquake Research Institute, University of Tokyo*, 43, 237–239.
- [20] Utsu, T. (1965). A method for determining the value of b in a formula $\log N = a - bM$ showing the magnitude-frequency relation for earthquakes. *Geophysical Bulletin, Hokkaido University*, 13, 99–103.
- [21] Wiemer, S., & Wyss, M. (2000). Minimum magnitude of completeness in earthquake catalogs: Examples from Alaska, the western United States, and Japan. *Bulletin of the Seismological Society of America*, 90(4), 859–869.
- [22] Gualandi, A., Nichele, C., Serpelloni, E., Ferretti, A., & Bürgmann, R. (2025). Pre-seismic b-value anomalies in subduction zones: A global survey. *Nature Geoscience*, 18, 112–120.

- [23] Segall, P., & Harris, R. (1987). Earthquake deformation cycle on the San Andreas fault near Parkfield, California. *Journal of Geophysical Research: Solid Earth*, 92(B10), 10511–10525.
- [24] Stauffer, D., & Aharony, A. (1994). *Introduction to Percolation Theory* (2nd ed.). Taylor & Francis, London.
- [25] Turcotte, D. L. (1992). *Fractals and Chaos in Geology and Geophysics*. Cambridge University Press, Cambridge.